

Weight perturbation across three AI weather models

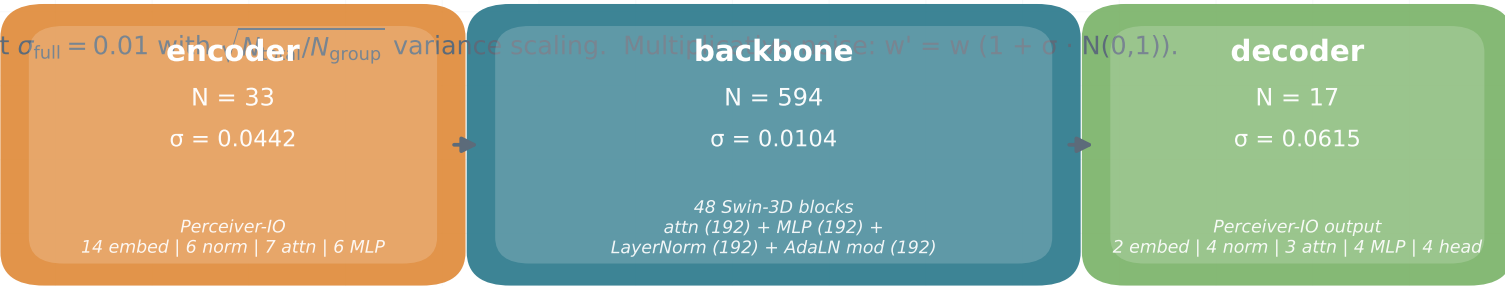


Phase 2 layer-group ablation at $\sigma_{full} = 0.01$ with $\sqrt{\sigma_{full}^2 / N_{group}}$ variance scaling. Multiplier: $w' = w(1 + \sigma \cdot N(0,1))$.

Aurora

Swin-Transformer 3D U-Net

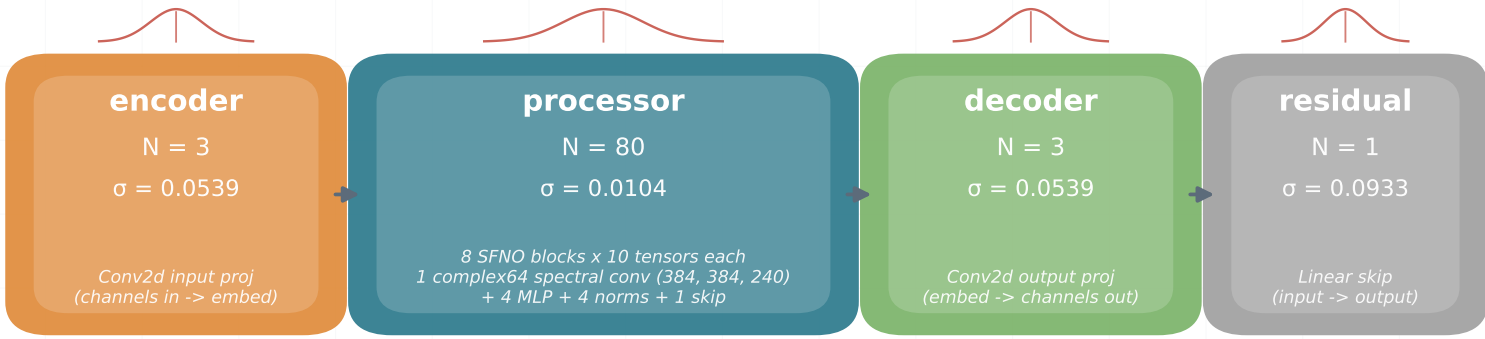
N_total = 644



SFNO

Spherical Fourier Neural Operator

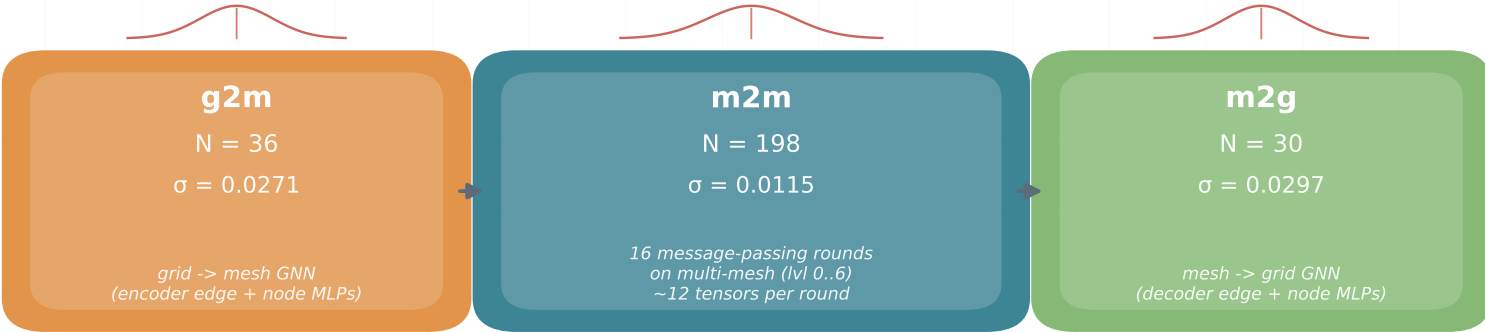
N_total = 87



GraphCast

Multi-Mesh Graph Neural Network

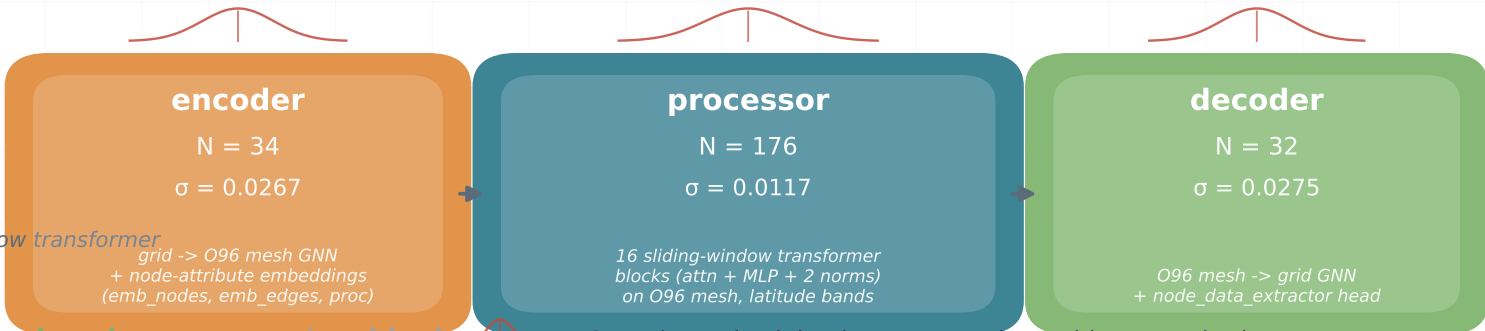
N_total = 264



AIFS

GNN encoder/decoder + sliding-window transformer

N_total = 242



encoder **processor** **decoder** **aux / residual** = Gaussian noise injection on every learnable tensor in the group